(c) $P(\mu-\sigma < X < \mu+\sigma) = P(-1 < Z < 1)$

$= 2\,P(0 < Z < 1)$

$= 2\,(0.3413)$

$= 0.6826$

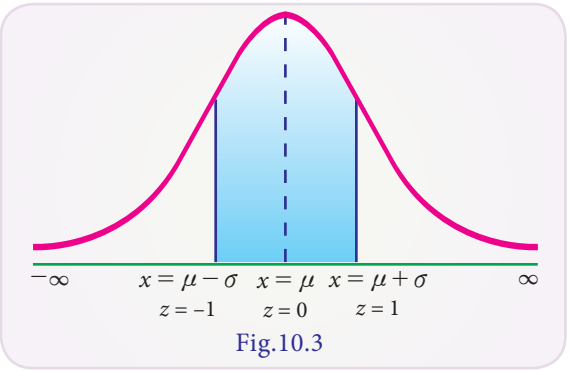


Fig.10.3

(d) $P(\mu-2\sigma < X < \mu+2\sigma)$

$= P(-2 < Z < 2)$

$= 2\,P(0 < Z < 2)$

$= 2\,(0.4772)$

$= 0.9544$

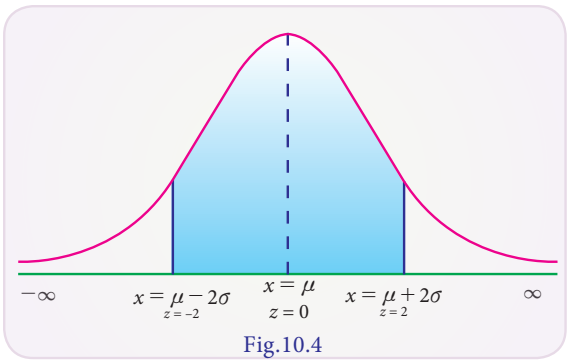


Fig.10.4

(e) $P(\mu-3\sigma < X < \mu+3\sigma)$

$= P(-3 < Z < 3)$

$= 2\,P(0 < Z < 3)$

$= 2\,(0.49865)$

$= 0.9973$

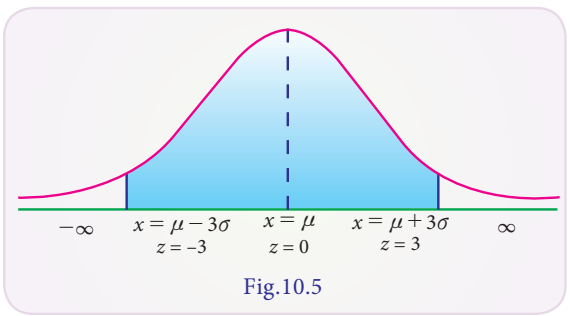


Fig.10.5

(f) $P(|X-\mu| > 3\sigma = P(|Z| > 3)$

$= 1 - P(|Z| < 3)$

$= 1 - P(-3 < Z < 3)$

$= 1 - 0.9973$

$= 0.0027$

Example 10.22

Find the area between $z = 0$ and $z = 1.56$

Solution:

$P(0 < Z < 1.56) = 0.4406$

(from the Normal Probability table)

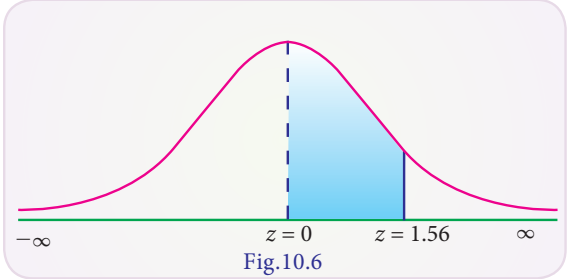


Fig.10.6